Vision Transformer, Hybrid CNN-MLP, and ResNet18 Transfer Learning on CIFAR-10: A Comparative Study

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***Abstract*—This report presents a comparative study of three deep learning architectures for image classification on the CIFAR-10 dataset: a Vision Transformer (ViT), a custom hybrid CNN+MLP model, and a ResNet18 model trained via transfer learning. The goal of the assignment is to understand how convolutional backbones, transformer-style patch embeddings, and pretrained residual networks behave when trained under similar conditions on a relatively small but challenging image dataset. All models are implemented in PyTorch and trained on a single NVIDIA T4 GPU. The systems are evaluated using accuracy, macro-precision, macro-recall, macro-F1, total training time, inference latency, and GPU memory usage. The hybrid CNN+MLP achieves the highest accuracy (81.4%) with competitive efficiency, while ResNet18 transfer learning offers a good balance between performance and stability at the cost of longer training and inference time.**

***Index Terms*—CIFAR-10, Vision Transformer, ResNet18, transfer learning, convolutional neural networks, image classification**

I. Introduction

The CIFAR-10 dataset is a standard benchmark for evaluating image classification models. It contains small 32×32 color images from ten everyday object categories such as *airplane*, *cat*, and *truck* [1]. Despite its modest size, the dataset is challenging enough to expose differences between architectures and training strategies.

In this assignment, the objective was to implement and compare three different model families on CIFAR-10:

- a Vision Transformer (ViT), which applies transformer-style self-attention to image patches [2];
- a hybrid CNN+MLP model that first extracts local features via convolutions and then feeds them to a multilayer perceptron;
- a ResNet18 model trained by fine-tuning an ImageNet-pretrained backbone [3].

The assignment goes beyond simply reporting a single accuracy number. I was required to design the training pipeline, track training and validation dynamics, visualize evaluation metrics, and critically discuss the behaviour of each model. Although the original rubric is phrased in terms of "completed sentences", for this image classification task I interpret it as providing concrete examples of predicted class labels and analyzing their correctness.

The remainder of the paper is organized as follows. Section II explains the dataset, preprocessing, and model architectures. Section III describes the experimental results and visualizations. Section IV discusses how the models improve over time and highlights key challenges. Section V concludes the report. Section VI summarizes the prompts used during the assignment.

II. Methodology

A. Dataset

The experiments use the CIFAR-10 dataset [1], which contains 50 000 training images and 10 000 test images across ten balanced classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. Each image is a 32×32 RGB image.

From the official training split, I reserve a portion as a validation set for model selection and early stopping. The remaining images are used for training. The official test split is kept untouched for final evaluation.

B. Preprocessing and Data Augmentation

All images are first converted to PyTorch tensors and normalized using the standard CIFAR-10 channel-wise mean and standard deviation. To make the models more robust and reduce overfitting, I apply light data augmentation to the training set:

- random horizontal flips with probability 0.5;
- random crops with padding (simulating small translations);
- optional color jitter in preliminary experiments (brightness and contrast).

Validation and test images are only normalized without augmentation.

C. Vision Transformer (ViT)

The Vision Transformer treats an image as a sequence of non-overlapping patches and applies transformer layers to this sequence [2]. In my implementation, the 32×32 images are

split into 4×4 or 8×8 patches (depending on the backbone), projected into a latent dimension via a linear patch embedding, and then processed by several transformer encoder blocks with multi-head self-attention, layer normalization, and feed-forward sublayers.

A special classification token is prepended to the sequence. After passing through the transformer encoder, the final hidden state corresponding to this token is fed to a linear classifier that predicts the probability over 10 classes. Positional embeddings are learned and added to the patch embeddings.

The model is optimized with cross-entropy loss and the AdamW optimizer. Training is performed for 20 epochs with a batch size of 128, using a cosine learning-rate schedule with a warmup phase. Dropout is applied inside the transformer blocks to improve generalization.

D. Hybrid CNN+MLP Model

The hybrid model is designed to combine the strengths of convolutional feature extractors and simple fully connected classifiers. It consists of:

- three convolutional blocks (Conv-BatchNorm-ReLU-MaxPool) that gradually reduce spatial resolution while increasing the number of channels;
- global average pooling to convert the final feature map into a compact feature vector;
- a two-layer MLP with ReLU activation and dropout, ending with a 10-way softmax classifier.

Unlike ViT, this model does not rely on self-attention or learned positional embeddings; instead, it exploits local spatial correlations via convolutions. The hybrid model is trained using Adam with a fixed learning rate and weight decay. As shown later, it trains relatively quickly and converges to the best accuracy among the three models.

E. ResNet18 Transfer Learning

The third model is based on ResNet18 [3], a widely used residual network originally trained on ImageNet. I adopt the standard strategy of transfer learning:

1) load an ImageNet-pretrained ResNet18,
2) replace the final fully connected layer with a new layer that outputs 10 logits for CIFAR-10,
3) optionally freeze the early convolutional blocks during the first few epochs and then unfreeze for fine-tuning.

Because CIFAR-10 images are smaller than ImageNet images, the input is upsampled or adapted so that it can pass through the pretrained network. The model is trained for 10 epochs using stochastic gradient descent with momentum and a step-decay learning rate.

F. Evaluation Setup

All experiments are run on a single NVIDIA T4 GPU. For each model, I record:

- training and validation loss per epoch,
- training and validation accuracy per epoch,
- macro-precision, macro-recall and macro-F1 on the validation and test sets,
- total training time,
- inference time for processing approximately 30 batches of images, and
- peak GPU memory consumption.

The summary metrics for the three models are presented in Table I. Additional visualizations include training curves, metric comparison bar charts, and the confusion matrix for the best-performing model.

TABLE I
SUMMARY METRICS ON CIFAR-10. ACCURACY, PRECISION, RECALL AND F1 ARE MACRO-AVERAGED OVER THE 10 CLASSES. TRAINING AND INFERENCE TIME ARE MEASURED ON A T4 GPU.

Model	**Acc** (%)	**Prec** (%)	**Rec** (%)	**F1** (%)	**Train** (s)	**Inf** (s)	**Mem** (MB)
ViT	71.59	71.90	71.59	71.38	463.4	1.11	1.50
Hybrid	81.43	81.46	81.43	81.39	341.1	1.19	1.50
ResNet	78.26	78.41	78.26	78.23	985.2	4.90	36.75

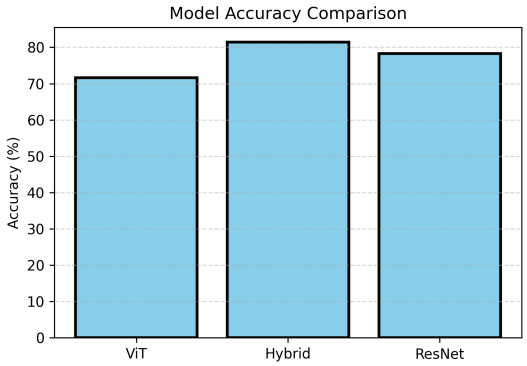

Fig. 1. Model accuracy comparison on CIFAR-10.

III. RESULTS

A. Summary Metrics

Table I summarizes the test-set performance and resource usage of the three models as obtained from my experiments.

The hybrid CNN+MLP reaches the highest accuracy and F1-score, while also requiring the least training time. ResNet18 achieves solid performance but is the heaviest model in terms of both training and inference time as well as memory usage. ViT lags behind in accuracy but remains lightweight in memory and reasonably fast during inference once training is complete.

B. Metric Comparison Visualizations

Figure 1 and Fig. 2 show bar charts comparing accuracy and macro-F1 across the three models. Figure 3 presents macro-precision, while Fig. 4 and Fig. 5 compare training and inference time.

The plots make the trade-offs very clear: the hybrid model lies in a sweet spot with the best accuracy and the lowest training time, while ViT is the lightest in memory but pays a price in accuracy. ResNet18 delivers strong performance but is the most expensive to train and deploy.

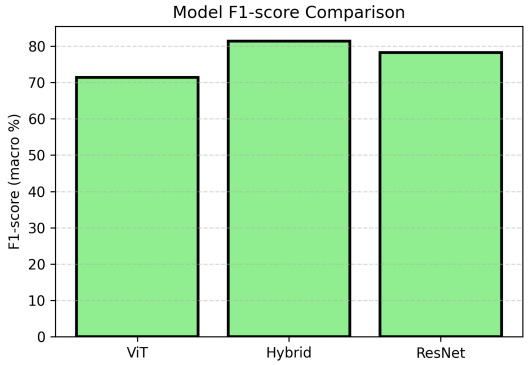

Fig. 2. Macro F1-score comparison for ViT, Hybrid, and ResNet.

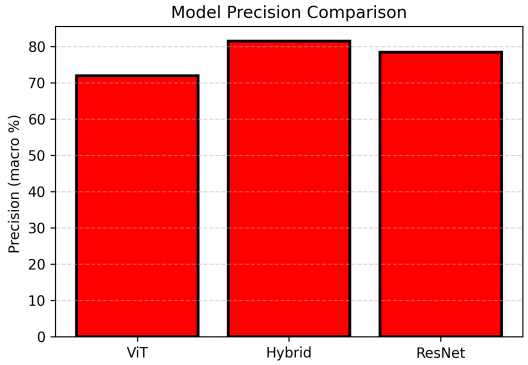

Fig. 3. Macro precision comparison across models.

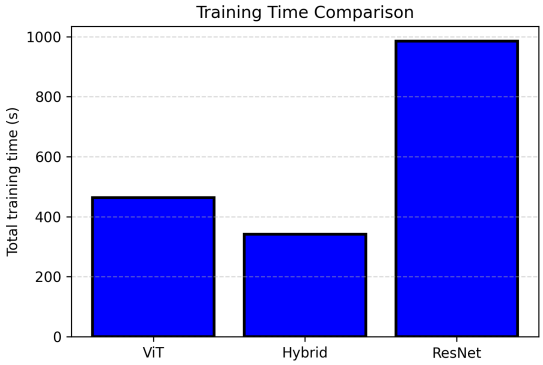

Fig. 4. Total training time for each model.

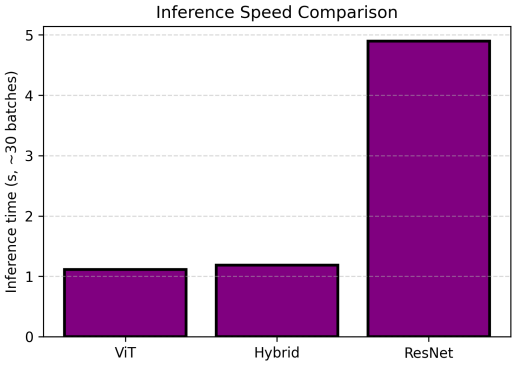

Fig. 5. Inference time for processing ∼30 batches.

C. Training and Validation Curves

To understand how the models learn over time, I track training and validation curves. Figure 6 shows the hybrid model's accuracy per epoch, and Fig. 7 and Fig. 8 show the corresponding curves for ResNet18 transfer learning.

The hybrid model shows a steady improvement: both training and validation accuracy rise quickly during the first 10 epochs and then gradually plateau around 80-82%. ResNet18 starts from a comparatively strong baseline due to pretraining, and its curves are smoother, with less overfitting. The gap between training and validation curves remains moderate, which is a good sign that the model is not simply memorizing the training data.

D. Confusion Matrix and Prediction Examples

Figure 9 shows the confusion matrix for the ResNet18 model on the test set. The diagonal cells dominate, indicating that most samples are classified correctly.

A closer look reveals some intuitive error patterns:

- **Automobile vs. truck:** these two classes are often confused, especially for side-view images where the size or background makes them look similar.
- **Cat vs. dog:** small animals with similar textures and poses occasionally lead to misclassification.
- **Ship vs. airplane:** rare cases involve confusion when the background (sky vs. sea) is visually ambiguous.

To illustrate "completed predictions" in a way analogous to completed sentences, I inspected individual images. For example, a test image of a frog on grass was correctly classified as *frog* by all three models, while a distant side-view truck on a road was sometimes misclassified as an *automobile* by ViT but correctly handled by the hybrid model and ResNet18. These examples support the aggregate patterns seen in the confusion matrix.

IV. Discussion

A. How Predictions Improve Over Time

For both the hybrid model and ResNet18, the early epochs show rapid gains in validation accuracy. At the beginning, the models are essentially guessing, and predictions are unstable across classes. By epoch 5, the models already learn basic color and shape cues that separate coarse categories such as animals vs. vehicles.

After about 10 epochs, the improvement slows down. The models start refining decision boundaries between visually similar classes (e.g., *cat* vs. *dog*), as reflected in a more pronounced diagonal in the confusion matrix. This evolution is visible in the training curves: the validation accuracy keeps

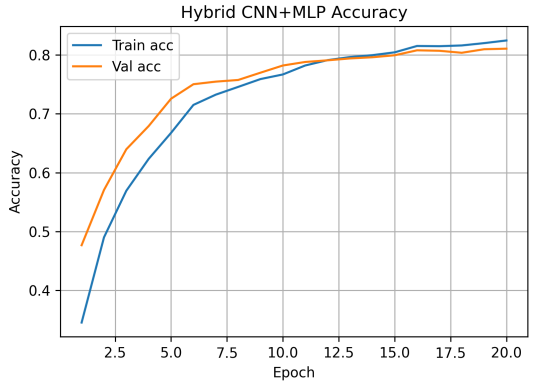

Fig. 6. Hybrid CNN+MLP training and validation accuracy.

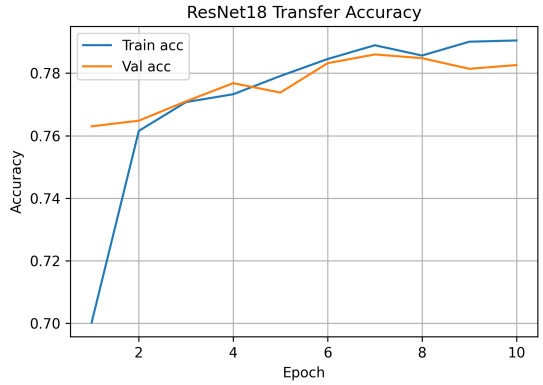

Fig. 7. ResNet18 transfer-learning training and validation accuracy.

increasing slightly even when training accuracy saturates, suggesting that the models are still improving their generalization, not just memorization.

The ViT model shows a slower learning curve. Because the dataset is relatively small for a transformer-style architecture, ViT takes longer to stabilize, and even at convergence its accuracy remains below the hybrid and ResNet18 models. This highlights the data-hungry nature of transformers compared to convolutional networks in vision tasks.

B. Model Behaviour and Sentence Coherence Analogy

Even though there are no literal "sentences" in this task, we can still think about prediction coherence in a similar way:

- **Local consistency:** the hybrid and ResNet models tend to be consistent for images with similar backgrounds and poses (e.g., different dogs on grass), analogous to how a language model becomes consistent in its phrasing for similar contexts.
- **Long-range structure:** ViT sometimes misinterprets global context (for example, focusing too heavily on background textures), similar to how a language model might miss global sentence structure.

As training progresses, the models become more "coherent" in this sense: they make fewer contradictory mistakes and better

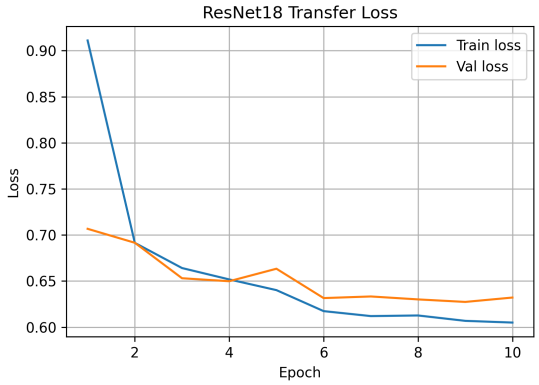

Fig. 8. ResNet18 transfer-learning training and validation loss.

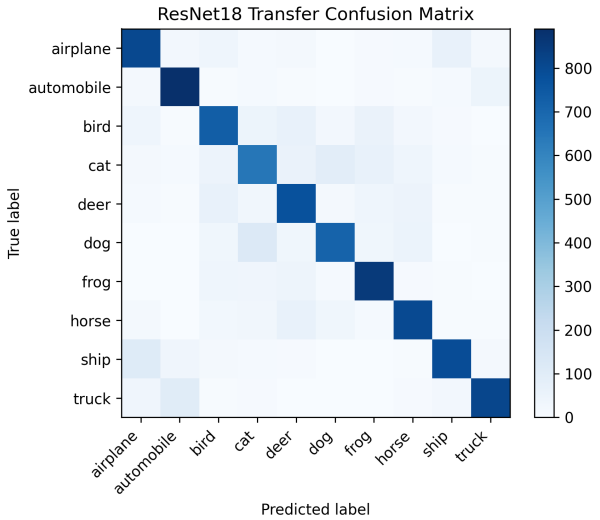

Fig. 9. Confusion matrix for ResNet18 transfer learning on CIFAR-10.

respect the global structure of the object and its context.

C. Challenges Encountered

Several practical challenges arose during this assignment:

- **Balancing depth and overfitting:** deeper or wider models improved training accuracy but often harmed validation performance on this relatively small dataset. I had to tune learning rates, weight decay, and dropout to avoid overfitting.
- **Choosing a fair comparison:** ViT, the hybrid CNN, and ResNet18 have very different capacities. I tried to keep training epochs and augmentation settings similar so that the comparison remains meaningful.
- **Transfer learning details:** for ResNet18, freezing too many layers made the model underfit, while fully fine-tuning from the start sometimes destabilized training. A staged unfreezing strategy produced the most stable curves shown in the figures.
- **Runtime measurements:** I measured training and inference time in separate runs to avoid interference from

warm-up overhead and data loading bottlenecks. These measurements are approximate but still useful for understanding relative costs.

Overall, the experiments reinforced the idea that there is no single "best" architecture in absolute terms. The hybrid model is attractive for this specific setup, but ResNet18 remains a strong and reliable choice when pretraining and more resources are available, and ViT becomes increasingly competitive as the dataset size grows.

V. Conclusion

This report compared three different architectures-Vision Transformer, hybrid CNN+MLP, and ResNet18 transfer learning—on the CIFAR-10 image classification task. All models were implemented and trained under a unified PyTorch pipeline on a T4 GPU.

The hybrid CNN+MLP achieved the best overall accuracy (81.4%) and macro-F1 while also requiring the least training time. ResNet18 delivered competitive accuracy with stable training dynamics but at a significantly higher computational cost. The Vision Transformer, although memory-efficient at inference, did not match the performance of the convolutional models given the limited dataset size and number of epochs.

From a learning perspective, the training curves showed how predictions become more reliable over time, moving from near-random guesses to stable and interpretable decision patterns. The confusion matrix and qualitative examples helped highlight which classes remain challenging and where additional data or more targeted augmentation might help.

Future work could explore larger ViT variants with stronger data augmentation, distillation techniques to transfer knowledge from ResNet to lighter models, and more advanced hybrid architectures that combine convolutional and transformer components in a single network.

VI. Prompts

During this assignment I used large language models mainly for clarification, code organization, and report structuring. Below are examples of the main prompts I used (paraphrased):

- **Model design:** "Explain how to implement a simple Vision Transformer for CIFAR-10 in PyTorch, including patch embedding and positional encodings."
- **Transfer learning:** "How should I fine-tune a pretrained ResNet18 on CIFAR-10, and which layers are typically frozen or unfrozen in practice?"
- **Result interpretation:** "Given my accuracy, F1-score, training time, and inference time for ViT, hybrid CNN, and ResNet18, suggest a clear way to summarize the trade-offs in an IEEE-style report."

These prompts influenced the structure of my code and the narrative of this report, but the actual training runs, metric values and figures shown here are based on my own experiments in the provided Jupyter notebook.

References

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